PROJECT PHOENIX X

AUTONOMOUS MULTI-SPECTRAL EDGE-AI SAR UAV | PHASE-1 PROTOTYPE BILL OF MATERIALS

Smart India Hackathon 2026 • **Problem Statement ID:** SIH26177 • **Sponsoring Organization:** Qualcomm Inc.
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Executive Summary: Problem statement SIH26177 (Qualcomm Inc.) mandates an autonomous tactical UAV engineered for emergency disaster reconnaissance in communications-blackout and GPS-compromised environments. To ensure rapid, scalable procurement by national and regional disaster authorities (NDRF/SDRF), Project Phoenix X implements a **calibrated two-tier BOM strategy**. The **Phase-1 Working Prototype (■37,400)** directly proves the dual-brain computing architecture featuring the **Arduino UNO Q (4GB LPDDR4 / 32GB eMMC)**, 250 Hz Zephyr RTOS flight loops, heterogeneous on-device AI inference on Qualcomm silicon (CPU/GPU/DSP), and 17-byte LoRa packet telemetry. The subsequent **Phase-2 Industrial Production Unit (■85,000 – ■1,20,000)** delivers enterprise-grade Search-and-Rescue capabilities at **1/10th the acquisition cost** of imported commercial platforms (e.g., DJI Matrice 30T at ■12–14 Lakhs) while maintaining 100% air-gapped data integrity.

1. Phase-1 Working Prototype Bill of Materials (Calibrated ■37,400)

No.	Subsystem / Component	Qty	Engineering Specification / Part Description	Unit (■)	Total (■)	Tactical Function & Operational Role
1	**Main Compute (Dual-Brain SBC)**	1	**Arduino UNO Q** (ABX00163, 4 GB RAM / 32 GB eMMC) • Qualcomm Dragonwing QRB2210 MPU (Debian Linux) • STM32U585 MCU (250 Hz Zephyr RTOS loop)	**8,800**	**8,800**	Single-board dual-brain partition: Heterogeneous AI (CPU/GPU/DSP) runs INT8 YOLO (~3.5W draw); STM32 handles deterministic flight control.
2	**Airframe & Propulsion Kit**	1 set	• F450 / S500 quadcopter composite frame • 4× A2212 1000KV BLDC motors • 4× 30A SimonK / BLHeli electronic speed controllers • 2 pairs 1045 nylon-carbon composite propellers	**4,800**	**4,800**	Delivers $\beta \geq 2.0$ thrust-to-weight ratio to lift 1.58 kg AUW platform (DGCA Micro-UAV compliant < 2.0 kg) and resist 12 m/s wind gusts.
3	**Avionics Sensor Suite & GNSS**	1 set	• u-blox NEO-M8N GPS + magnetic compass mast • MPU6050 6-DOF IMU (accel/gyro) • MS5611 high-precision barometric altimeter • ST VL53L1X 1D Time-of-Flight (ToF) downward laser	**4,200**	**4,200**	Multi-sensor fusion for EKF state estimation; downward ToF feeds real-time H_AGL to Tait-Bryan ground ray-casting projection.
4	**Power Subsystem & Balance Charger**	1 set	• 3S 11.1V 4500mAh 30C LiPo battery pack • iMAX B6 80W digital balance charger • 12V 5A regulated AC/DC power supply	**4,000**	**4,000**	Provides 25–28 minutes search endurance per sortie; adheres to Bouabdallah 38% battery mass fraction sizing law.
5	**Manual RC Safety Failover Link**	1 set	• FlySky FS-i6 6-channel 2.4 GHz AFHDS transmitter • FS-iA6B 6-channel receiver (iBus / PPM protocol)	**3,900**	**3,900**	Emergency pilot takeover channel for field safety redundancy and DGCA fail-safe override compliance.
6	**Optical Vision (RGB Camera)**	1	• 1080p Full HD UVC USB camera module • OmniVision OV2710 sensor, 120° wide-angle M12 lens	**1,100**	**1,100**	Daylight stream for survivor clothing/gesture recognition and visual identification of dangling power cables & hazards.
7	**Thermal Infrared Camera (LWIR)**	1	• Melexis MLX90640 radiometric thermal array • 32×24 pixels, 8–14 µm LWIR, 55°×35° FOV, I2C	**3,400**	**3,400**	24/7 night, fog, and smoke penetration; isolates 37°C survivor body heat anomalies at rubble/mud interfaces.
8	**Long-Range RF Telemetry (LoRa)**	2	• Semtech SX1262 (Ebyte E22-900T22S) • 868 MHz band (Indian WPC de-licensed), +22 dBm, SPI (1× on UAV, 1× ground USB station bridge)	**850**	**1,700**	Broadcasts 17-byte binary bit-packed triage telemetry over 3–5+ km NLoS range during complete cellular/Wi-Fi blackouts [2, 3].
9	**Power Distribution & Integration Gear**	1 lot	• Matek PDB with dual 5V/12V BECs • 5V 3A low-noise buck converter (for Arduino UNO Q) • Silicone anti-vibration damping ball gimbal mount • 14 AWG wiring, XT60 connectors, landing skids, hardware	**5,500**	**5,500**	Harmonic vibration isolation protecting flight IMU; clean, brownout-free regulated power rail to Qualcomm MPU mission board.
	TOTAL PROTOTYPE INVESTMENT	**9 Lots**	**Complete Functional Hardware Prototype (Evaluation Ready)**		**■37,400**	**Calibrated end-to-end autonomous SAR platform**

2. Engineering Validation & Architectural Integrity

Subsystem Domain	Engineering Trade-Off & Empirical Validation	Operational Outcome in Phoenix X
Memory & Storage (4 GB / 32 GB UNO Q)	Flight control runs on STM32U585's dedicated 786 KB SRAM. The Qualcomm QRB2210 Linux MPU utilizes ~200 MB (OS), ~180 MB (INT8 YOLO buffers), ~120 MB (OpenCV), and ~120 MB (Ray-casting & LoRa), totaling **~620 MB active RAM**. The 32 GB eMMC holds 4.2 GB (OS), 12 GB (offline GIS vector tiles), and 8.5 GB (black-box flight logs).	**85% Free Headroom:** Eliminates kernel panics; vibration-proof eMMC prevents micro-SD card corruptions under motor harmonics.
Power Budget (~5.5W Electronics)	Propulsion draws ~168W at hover (4× 42W). Onboard electronics draw ~5.5W under peak AI workload (UNO Q ~3.5W, cameras ~0.9W, sensors/GNSS ~0.4W, LoRa burst ~0.5W, regulators ~0.6W).	**>96% Power for Propulsion:** Electronics consume <4% of battery envelope, preserving maximum endurance for search sorties.
Mass Budget (1.58 kg AUW)	Component breakdown: Frame (282g), 4× Motors (208g), ESCs (96g), Props (44g), LiPo (362g), UNO Q (42g), Cameras/Mount (79g), GNSS/ToF/Sensors (48g), LoRa/RC (33g), PDB/Wiring (216g), Reserve (170g).	**DGCA Micro Compliant:** Stays under 2.0 kg threshold with a 170g unallocated payload reserve for droppable beacons.
RF Telemetry (Semtech 868 MHz)	Under Government of India WPC / DoT GSR 564(E), the 865–867 MHz band is officially de-licensed up to 1 Watt (+30 dBm). Unlike 433 MHz (congested by remotes) or 2.4 GHz (blocked by concrete/foliage), 868 MHz penetrates deep urban collapse voids.	**3–5+ km NLoS Link:** Sustains resilient 17-byte survivor triage telemetry [2, 3] without cellular infrastructure.

3. Roadmap: Phase-1 Prototype vs. Phase-2 Production vs. Commercial Alternatives

Subsystem Parameter	Phase-1 Hackathon Prototype	Phase-2 NDRF Production Unit	Imported Enterprise Alternative
Unit Acquisition Cost	■**37,400** (Calibrated Build)	■**85,000 – ■1,20,000 (~■1.2 Lakh)**	■**12,00,000 – ■14,00,000** (DJI M30T)
Airframe Structure	F450 Glass-Fiber / Nylon Composite	Toray 3K Foldable Carbon-Fiber Monocoque	Proprietary Magnesium-Aluminum Shell
Thermal Imaging Sensor	MLX90640 (32×24 Radiometric Array)	FLIR Lepton 3.5 (160×120) or InfiRay LWIR	Proprietary 640×512 Radiometric Sensor
Geodetic Positioning	u-blox NEO-M8N Multi-GNSS (2.5m CEP)	u-blox ZED-F9P Centimeter RTK-GNSS	Proprietary RTK Module (+■2 Lakh add-on)
All-Weather Sealing	Open prototype development chassis	IP54 Conformal-Coated Sealed Enclosure	IP55 Weather Sealing
Data Sovereignty & Cyber	100% Air-Gapped (Zero Cloud Uplink)	**100% Air-Gapped, HMAC-SHA256 Encrypted**	Foreign cloud infrastructure dependencies
Tactical Fleet Value	Rapid proof-of-concept verification	**Deploy 10 Phoenix UAVs for cost of 1 M30T**	Single-point-of-failure budget risk

Certification & Compliance Note: All components listed in the Phase-1 BOM are commercially off-the-shelf in India. The all-up-weight of 1.58 kg complies with DGCA Micro-Category guidelines (< 2.0 kg), eliminating heavy remote pilot certification hurdles for rapid hackathon flight testing. RF communications strictly adhere to WPC GSR 564(E) unlicensed 865–867 MHz frequency allocations.